

Theming & Internationalization

▼ Relevant source files

Frontend/STUDER/src/app/app.routes.ts

Frontend/STUDER/src/app/core/i18n/i18n.provider.ts

Frontend/STUDER/src/app/core/i18n/language.interceptor.ts

Frontend/STUDER/src/app/core/i18n/language.service.ts

Frontend/STUDER/src/app/core/theme/theme.service.ts

Frontend/STUDER/src/app/shared/components/rich-text/entity-popover/entity-popover.component.ts

Frontend/STUDER/src/assets/i18n/en.json

Frontend/STUDER/src/assets/i18n/es.json

Frontend/STUDER/src/assets/images/back-dark.png

Frontend/STUDER/src/assets/images/back-light.png

Frontend/STUDER/src/styles.css

This section details the mechanisms used to provide a multi-language, themeable user experience in the STUDER platform. The system leverages Angular's reactive patterns, CSS custom properties (variables), and the `ngx-translate` ecosystem to manage visual appearance and localized content.

Theming System

The STUDER frontend implements a theme switching mechanism primarily driven by CSS variables and a centralized `ThemeService`. The application supports two primary modes: `light` (default) and `dark`.

Implementation Details

The theming logic is encapsulated in the `ThemeService`

Frontend/STUDER/src/app/core/theme/theme.service.ts

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It manages the persistence of the

user's preference in `localStorage` and manipulates the `document.body` class list to trigger CSS transitions.

- **Persistence:** Uses the key `preferred_theme`
`Frontend/STUDER/src/app/core/theme/theme.service.ts` 7
- **Application:** Themes are applied by adding `theme-light` or `theme-dark` classes to the `<body>` element `Frontend/STUDER/src/app/core/theme/theme.service.ts` 10-11

CSS Variable Architecture

The global styles define a `:root` pseudo-class containing default variables, which are then overridden by specific body class selectors `Frontend/STUDER/src/styles.css` 36-156

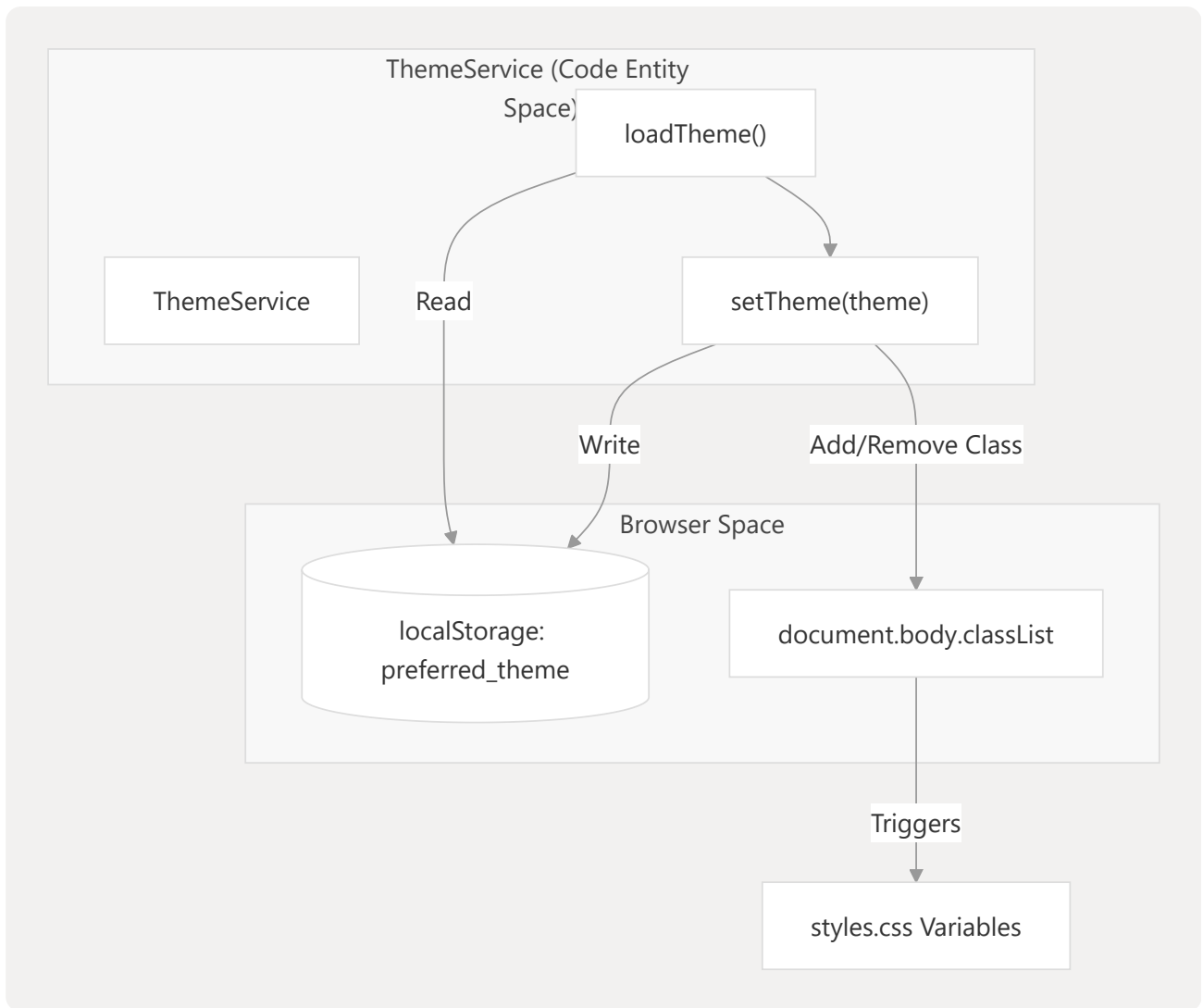
Variable Category	Light Mode Value (Default)	Dark Mode Value
<code>--color-bg</code>	<code>#fff</code> <code>Frontend/STUDER/src/styles.css</code> 50	<code>#181818</code> <code>Frontend/STUDER/src/styles.css</code> 125
<code>--color-text-prim</code>	<code>#171717</code> <code>Frontend/STUDER/src/styles.css</code> 52	<code>#f5f5f5</code> <code>Frontend/STUDER/src/styles.css</code> 126
<code>--sidebar-bg</code>	<code>#f4f4f5</code> <code>Frontend/STUDER/src/styles.css</code> 66	<code>#1e1e22</code> <code>Frontend/STUDER/src/styles.css</code> 132
<code>--input-background</code>	<code>rgba(230, 230, 230, 0.41)</code> <code>Frontend/STUDER/src/styles.css</code> 61	<code>rgba(50, 50, 50, 0.6)</code> <code>Frontend/STUDER/src/styles.css</code> 140

Background Images

The system swaps background assets based on the active theme class:

- **Light:** `assets/images/back-light.png` `Frontend/STUDER/src/styles.css` 31
- **Dark:** `assets/images/back-dark.png` `Frontend/STUDER/src/styles.css` 27

Theme State Management Diagram



Sources: Frontend/STUDER/src/app/core/theme/theme.service.ts 1-23

Frontend/STUDER/src/styles.css 26-32

Internationalization (i18n)

STUDER uses `ngx-translate` for runtime translation and a custom `LanguageService` to manage locale state.

Configuration and Provider

The i18n system is initialized via the `provideI18n` function in the application configuration

Frontend/STUDER/src/app/core/i18n/i18n.provider.ts 5-12

- **Loader:** Uses `HttpClient` to fetch JSON files from `/assets/i18n/`
Frontend/STUDER/src/app/core/i18n/i18n.provider.ts 11
- **Fallback:** Defaults to Spanish (`es`) if a translation is missing
Frontend/STUDER/src/app/core/i18n/i18n.provider.ts 8

LanguageService

The `LanguageService` handles the business logic for switching languages and initializing the preferred locale
Frontend/STUDER/src/app/core/i18n/language.service.ts 8-56

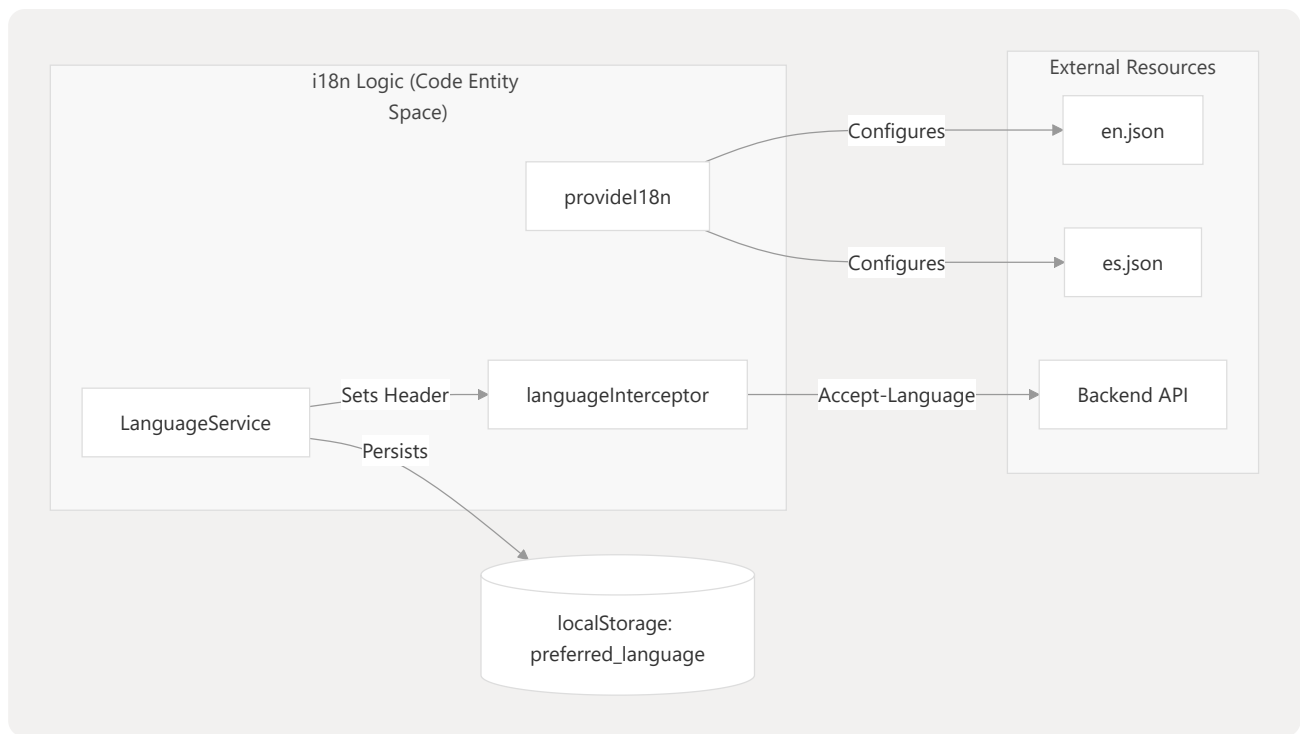
- **Initialization:** On startup, it checks `localStorage` for `preferred_language`. If empty, it detects the browser language via `navigator.language`
Frontend/STUDER/src/app/core/i18n/language.service.ts 27-35
- **Reactivity:** Exposes the current language via a `BehaviorSubject`
Frontend/STUDER/src/app/core/i18n/language.service.ts 10

Language Interceptor

To ensure the backend provides localized error messages or data, the `languageInterceptor` attaches the `Accept-Language` header to every outgoing HTTP request

Frontend/STUDER/src/app/core/i18n/language.interceptor.ts 3-6

i18n Data Flow Diagram



Sources: Frontend/STUDER/src/app/core/i18n/language.service.ts 1-56

Frontend/STUDER/src/app/core/i18n/language.interceptor.ts 1-6

Frontend/STUDER/src/app/core/i18n/i18n.provider.ts 1-12

Translation Schema

Translations are stored in structured JSON files. Both **en.json** and **es.json** follow a strict hierarchical schema to allow easy access via dot notation (e.g., `{{ 'login.title' | translate }}`).

Key Modules in Schema

- **common** : Global terms like "Hello", "Language", "Theme"
Frontend/STUDER/src/assets/i18n/en.json 2-6
- **landing** : Marketing slogans and feature descriptions for the landing page
Frontend/STUDER/src/assets/i18n/en.json 46-85
- **discussions** : UI text for the discussion forums, including form placeholders and status labels
Frontend/STUDER/src/assets/i18n/en.json 148-173

- **profile** : Account management and user profile actions

Frontend/STUDER/src/assets/i18n/en.json 122-135

Rich Text Integration

The internationalization system works alongside the **RichTextParserService** and

EntityPopoverComponent . While the UI labels are translated via JSON files, the dynamic entities (users, tags, courses) use color variables defined in the theme CSS

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Entity Type	Prefix	CSS Variable	Default Color
User	@	--entity-user-color	#4285f4 (Blue)
Tag	#	--entity-tag-color	#9c27b0 (Purple)
Course	&	--entity-course-color	#00bf63 (Green)
Contest	\$	--entity-contest-color	#ff3131 (Red)
Block	%	--entity-block-color	#ff9800 (Orange)

Sources: Frontend/STUDER/src/assets/i18n/en.json 1-174

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